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Improving Business Processes with Business Process Modelling Notation and Business Process Execution Language: An Action Research Approach

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Abstract

The first purpose of this paper is to improve business processes in an industrial firm with Business process modelling Notation (BPMN) and Business Process Execution Language (BPEL). The second purpose is to provide a scientific approach for bringing business analysts and IT professionals together in a framework of an Action Research. The research is conducted in one of the biggest distribution companies in Iran. Business process modelling Notation is applied to model As-Is and to-be situations of sale and distribution process. Both As-Is and To-Be models were simulated to compare their performance indexes. Moreover, models were implemented using Business Process Execution Language, so that the model can be used for automating and improving the process. In this study, Action Research methodology was used to find a resolution of an organizational issue with those who experience this issue directly and to improve scientific knowledge and real-life contexts simultaneously.

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Key words Business Process Reengineering, Business process modelling, Business process modelling Notation, Action Research, Business Process Execution Language

1. Introduction

Nowadays the ability of organizations to understand and react appropriately to the business events is a key to maintain competitiveness. The necessity of understanding, analyzing and effectively responding to constant changes in the market and risks, is the main reason for special attention paid to business processes (Kuster, et al., 2006). Using business processes modelling and changing processes into computer systems, can be a good answer to the environment complexity and changes (Rodriguez et al., 2007). In order to do so, more flexibility in business modelling as well as the information technology as an enabler and supporter is so important (Fernandez, et al., 2009). On the business side, it seems necessary to eliminate linear models and to move towards network models of business. On the business level, changing and improving processes and on the IT level, web services and using service-oriented architecture are the main cores of new technologies (Zimmermann *et al.*, 2003). Given the existing competition between modern organizations, facilitating business modelling and updating processes are hot topics. Because of the increase in business process management projects, the demand for business process modelling techniques has increased dramatically (Cull and Eldabi, 2010). BPMN and BPEL are two applicable tools for business process modelling. For instance, BPEL is used widely to specify and execute composite business processes (Kareliotis *et al.*, 2009). What is still considered an issue in today's businesses, is the ability to standardize some methods for decreasing the distance between industries and technology and to create agility in organizations. Environmental and technological changes can be suitable opportunities for organizations in order to have more access to resources and profitability.

This paper presented a business process modelling approach based on Action Research method to support practitioners in the field of BPR to choose appropriate best model to improve processes. It is important to emphasize that this work supports business process redesign where process improvement is the goal.

2. Related literature

2.1. Action Research (AR)

Action Research has been considered a distinguishing form of research since the 1940s. Kurt Lewin is generally regarded as one of its pioneers, applying AR to deal with social problems (Kock *et al.*, 1996). There are several definitions for Action Research but the main idea of AR is that it uses a scientific approach to study the resolution of important organizational or social issues together with those who experience these issues directly (Coughlan and Coughlan, 2002). AR tries to integrate scientific knowledge and real-life to make actions more effective and simultaneously build up a body of scientific knowledge. Previous research has shown that AR is particularly an appropriate method for information systems' researches (Mansell, 2008).

AR refers to the conjunction of three elements: Research, Action and Participation. It works through a cyclical four-step process of deliberately and consciously: Planning, Acting, Observing (or Evaluating) and Reflecting.

2.2 Business Process Reengineering (BPR)

“Reengineering” has been the most well-known and the most controversial management theory during recent years. It contradicts Adam Smith’s well-known principle of work division which has been around for centuries. Reengineering eliminates traditional task-based organizations. It does not accept gradual changes, but just deals with fundamental changes, taking rapid technological and economic changes into account. Successful BPR leads to efficiency improvement, customer-centricity and readiness for global competition (Tarokh *et al.*, 2008). The results of studying top experiences in reengineering consists of 29 rules, out of which 10 rules are listed in table 1, because they have been frequently used in literature and case studies (Reijers and Limam, 2005). The order of these rules may change in the future based on new experiences.

Table 1 most popular best practices in business process redesign

Best Practice	Definition
Task Elimination	Eliminate unnecessary tasks from a business process.
Task Composition	Combine small tasks into composite tasks and divide large tasks into workable smaller tasks
Integral Technology	Try to elevate physical constraints in a business process by applying new technology
Empower	Give workers the most decision-making authority and reduce middle

	managers
Order assignment	Let workers perform as many steps as possible for single orders
Re-sequencing	Move tasks to more appropriate places
Specialist-generalist	Consider to make resources more specialized or more general
Integration	Consider the integration with a business process of the customer or a supplier
Parallelism	Consider whether tasks may be executed in parallel
Numerical involvement	Minimize the number of departments, groups and persons involved in a business process

Evidence has shown that BPR implementation is a risky process (Motwani *et al.*, 2008). In order to minimize the risks of its implementation, organizations must investigate some critical success factors and assess their readiness for implementation (Abdolvand *et al.*, 2008). For instance, IT infrastructure is an important factor for success of BPR implementation (Eardley *et al.*, 2008)

2.3. Business Process Modelling

When Business Process Modelling (BPM) was introduced as a consulting project for modern organizations, it was often impractical and weakly designed for organizations (Gunasekaran and Kobu, 2002). Business process modelling even could not decide what to call itself. What happened to such a descriptive entity namely “workflow”? Supposedly, it was replaced by Business process modelling. The term “workflow” is now obsolete and forgotten (Havey, 2005). It’s difficult to guess what is now called “Business Process Modelling” was once called workflow. Many people think that “Business Processes Modelling” and “workflow” have different meanings; but “Business Processes Modelling” is beyond workflow and in fact made it outdated (Cremers, 2007). Today, business process management projects have increased severely and therefore the demand for business process modelling (BPM) techniques have augmented (Cull and Eldabi, 2010).

2.4 Classification of business processes modelling methods

Generally, business process (BP) models can be divided into three categories: Diagrammatic models, Mathematical models, and Business Process Languages (Vergidis *et al.*, 2008). Diagrammatic models are suitable for business process models that are presented as visual and graphical plans, like flowcharts. Their main application was in developing software systems. These models are appropriate for rapid and informal design and presenting of processes. They are often used without standards notations (Rigby and Bilodeau, 2005). These notations have lack of necessary meaning in expressing complicated and standard structures. Mathematical models are solid and complicated. These models are suitable for describing models and finding out their solidness and characteristics. One of the well-known models of this kind is Petri Net, which is a combination of mathematical and diagrammatic models. The third category, namely Business Process Languages consists of languages based on software and usually based on XML which supports modelling of business processes and in most cases executes processes (Wohed *et al.*, 2006). Fig. 1 shows the Venn diagram of these categories.

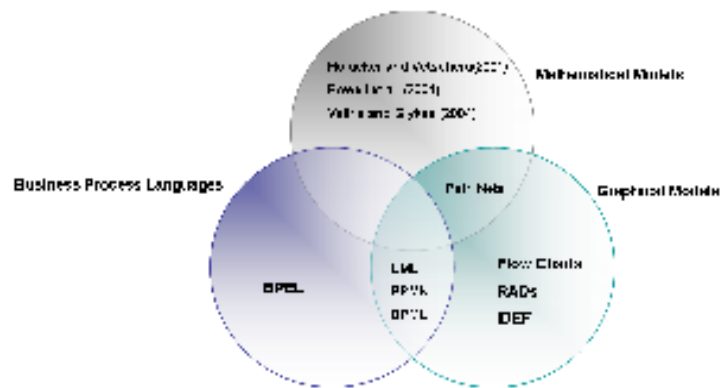


Fig. 1. The classification of business process modelling (Vergidis *et al.*, 2008)

Fig. 2 shows the relationships between some modelling standards and also the relationships between diagrammatic, mathematical and executive models.

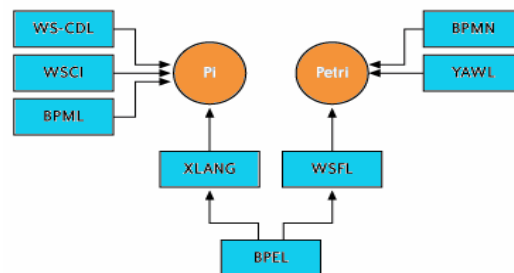


Fig. 2. The relationships between some modelling standards (Nissen, 1998)

2.5 Standards of Business Processes Modelling

Nowadays Service Oriented Architecture (SOA) becomes a new approach in organizational architecture, and web services are being used to build enterprise information based on this paradigm (Mateescu and Rampacek, 2008). The successful use of SOA depends on its symbiosis with business process management (BPM). An important part of BPM is usage of process modelling (Kruczynski, 2010). There are many standards used for modelling business processes. In this study Business Process Modelling Notation (BPMN) and Business Process Execution Language (BPEL) are used to improve company's processes. The former is presented by OMG and BPMI and the latter which is an industrial standard is presented by OASIS Consortium and BEA, Microsoft and IBM (IBM Corporation, 2005). In the existing situation, translating BPMN models into BPEL code is an important pace toward standards-based business process development environments (Ouyang *et al.*, 2008).

2.5.1 BPMN

Business Process Management Notation provides business analysts and system architects with some privilege level. Based on the latest statistics, BPMN is supported by more than 30 tools (Reijers and Limam, 2005). In fact, BPMN uses business process diagrams and creates a standard to capture business processes in early phases of systems development (Dijkman *et al.*, 2008). Fig. 3 shows a subset of BPMN elements. These diagrams consist of objects and sequence flow. Sequence flow shows the connection between two objects. An object can be an event, a task, or a gateway. An event can be the command for beginning a process, ending a process, instant end of a process, a received message, or a specific time which the process gets at. A task is a single activity which is waiting for execution during a process (Kuster, *et al.*, 2006; Havey, 2005).

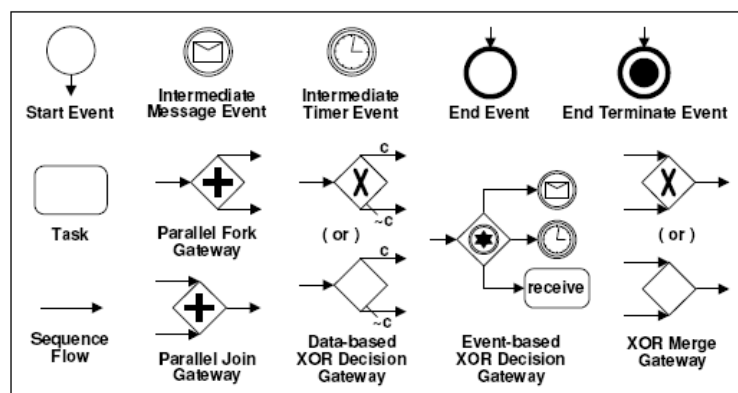


Fig. 3. Subset of BPMN elements (Ouyang *et al.*, 2006)

A Gateway is a routing structure which is used to control the convergence and divergence of a sequence flow. Gateway types are as follows:

- Parallel Fork Gateway: to provide concurrency in sequence flows
- Parallel Join Gateway: To provide concurrency in concurrent sequence flows
- Data/Event-based XOR Decision Gateway: To select one of the proposed sequence flow sets where the selection is based on process data or external event
- XOR Merge Gateway: to combine a set of proposed sequence flows and to form a sequence flow (Reijers and Limam, 2005)

BPMN features can be divided into 4 categories:

1. Flow Objects: Similar to events, activities and gateways, they are the basic elements for creating BP diagrams
2. Connecting Objects: Are used to connect flow objects and can be different types of arcs
3. Swim lanes: Are used to group activities (e.g. different roles or organizational units)
4. Artifacts: May be added to show information related to diagrams (Cremers, 2007)

2.5.2 BPEL

Business Process Execution Language for web services is a standard for implementing business processes on web service technology and this technology aggregates encapsulated functionalities to define high-value web services (Amme, 2009). Several execution platforms support the execution of BPEL processes. Some of these platforms provide graphical editing tools for defining BPEL processes. So these tools follow the BPEL syntax. Since these tools do not increase the abstraction level, they will not be useful during different phases of analysis and design of the development cycle (Recker and Mendling, 2005). In fact, BPEL is a language with block structure supported by different executive databases (Havey, 2005). For an amateur, Business processes modelling execution language (BPMEL) for web services is a completely technical computer term. Generally, making notation for BPMEL for web services can be interpreted as a formal definition of grammars which provides definition of behavior and interactions of an instance of a process with its partners and resources. A BPEL process can include structured activities. These structures can be easily determined. Fig. 4 shows an example of this determination. It demonstrates a purchase order process as BPEL.

```

001 <process name="purchaseOrderProcess"
002   targetNamespace="..."
003   xmlns="..."
004   xmlns:Ins="...">
...
044 <sequence>
045   <receive partnerLink="purchasing"
046     portType="Ins:purchaseOrderPT"
047     operation="sendPurchaseOrder"
048     variable="PO">
049   </receive>
050   <flow>
051     <links>
052       <link name="ship-to-invoice"/>
053       <link name="ship-to-scheduling"/>
054     </links>
055     <sequence>
056       <assign>
057         <copy>
058           <from variable="PO" part="customerInfo"/>
059           <to variable="shippingRequest"
060             part="customerInfo"/>
061         </copy>
062       </assign>
063       <invoke partnerLink="shipping"
064         portType="Ins:shippingPT"
065         operation="requestShipping"
066         inputVariable="shippingRequest"
067         outputVariable="shippingInfo">
068         <source linkName="ship-to-invoice"/>
069       </invoke>
070       <receive partnerLink="shipping"
071         portType="Ins:shippingCallbackPT"
072         operation="sendSchedule"
073         variable="shippingSchedule">
074         <source linkName="ship-to-scheduling"/>
075       </receive>
076     </sequence>
077   </sequence>
078   <sequence>
079     <invoke partnerLink="invoicing"
080       portType="Ins:computePricePT"
081       operation="initiatePriceCalculation"
082       inputVariable="PO">
083     </invoke>
084     <sequence>
085       <receive partnerLink="invoicing"
086         portType="Ins:computePricePT"
087         operation="sendShippingPrice"
088         inputVariable="shippingInfo">
089       <target linkName="ship-to-invoice"/>
090     </receive>
091     <sequence>
092       <receive partnerLink="invoicing"
093         portType="Ins:invoiceCallbackPT"
094         operation="sendInvoice"
095         variable="Invoice"/>
096     </sequence>
097   </sequence>
098   <sequence>
099     <invoke partnerLink="scheduling"
100       portType="Ins:schedulingPT"
101       operation="requestProductionScheduling"
102       inputVariable="PO">
103     </invoke>
104     <sequence>
105       <invoke partnerLink="scheduling"
106         portType="Ins:schedulingPT"
107         operation="sendShippingSchedule"
108         inputVariable="shippingSchedule">
109       <target linkName="ship-to-scheduling"/>
110     </invoke>
111   </sequence>
112 </flow>
113 </process>

```

Fig. 4. An example of BPEL purchase order

Business Processes Execution Language (BPEL) for web services provides a standard language for describing business processes consisting of functions defined through web service interfaces. This language includes design and execution application (Kanjilal, 2006). At the time of designing, modelling and development, tools can use exporting and importing BPMEL for web services in order to allow business analysts to determine processes. Also they allow designers to correct them and to bind process phases to specific service implementations (Fig. 5). The Workflow engine can use BPMEL for web services to control execution of processes and assignment of required services for implementation (Cremers, 2007; Wang *et al.*, 2006). Fig. 5 shows the distinction of a business process and its implementation.

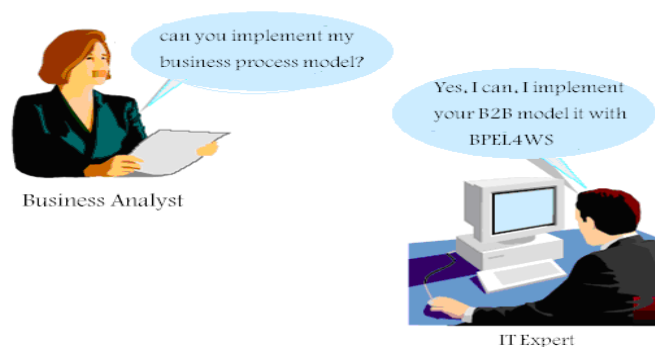


Fig. 5. Business analyst and IT expert negotiation

BPEL for web services supports a combination of complicated processes along with structure and relationships of different activities for creating a business process (BP). Business process can

be a web service (Hall, and Harmon, 2005). The important point in using BPMN and BPEL is that the user should have a clear understanding of structural differences between them: Firstly, BPMN in the cycle of its usage in BP management is used differently than BPEL. BPMN is useful during design and improvement of BP, whereas BPEL is useful during making the process model executable. Each of them has its own requirements. Secondly, BPMN is used by business analysts, whereas BPEL is used by technical analysts and software programmers. Each of them focuses on a different matter (Dikmans, 2008).

2.6 Future languages

The number of languages has been increased dramatically for process definition. For instance, Activity diagrams in UML 2.0. New languages like Business Process Execution Language (BPEL) and Business process modelling Notation (BPMN) have been developed and rapidly attracted people. The common point of these three languages is their emphasis on providing a complete and unified set of notations to model processes (Hall, and Harmon, 2005).

3. Research methodology

This research is done in one of the biggest distribution companies in Iran, Behpakhsh Company, to improve its processes. Firstly, company's processes were identified and a plan to change some processes was developed. Then the models of As-Is and To-Be were simulated and the new plan was applied in a small department of the company. The results of the new plan were observed and were presented to the manager and based on his attitudes, the proposed model for future situation was created.

In this study, Action Research methodology has been used to find a resolution of an organizational issue with those who have experienced this issue directly and also has been used for integrating scientific knowledge with real-life.

3.1 What is Action Research?

Action research is significantly different from traditional research in several ways. It is a process in which participants examine their own practice, systematically and carefully, using the techniques of research.

There are numerous definitions for AR, but all of them agree on the point that AR integrates research and real-life in order to improve scientific knowledge and real life contexts simultaneously. This method tries to describe academic assumptions of the researcher and his project (Altrichter *et al.*, 2002). AR is a participatory, democratic process concerned with developing practical knowledge in the pursuit of worthwhile human purposes, grounded in a participatory worldview which is emerging at this era (Brydon-Miller and Greenwood, 2003). As a research method, Action Research is cyclical. It assumes that understandings and actions emerge in a constant cycle. "Action research involves improvement of practice, understanding of practice and the situations in which practice occurs". (Kemmis and McTaggart, 2000)

According to Grundy and Kemmis there are three conditions necessary for AR to exist:

- A project takes as its subject matter a social or organizational practice, regarding it as a strategic action susceptible to improvement.
- The project proceeds through a spiral cycles of planning, acting, observing and reflecting with each of these activities being systematically and self-critically implemented and interrelated; and
- The project involves those who are responsible for the practice in each moments of the activity, widening participation in the project gradually to include others affected by the practice and maintaining collaborative control of the process.

A simple diagram of action-research spiral is shown in Fig. 6.

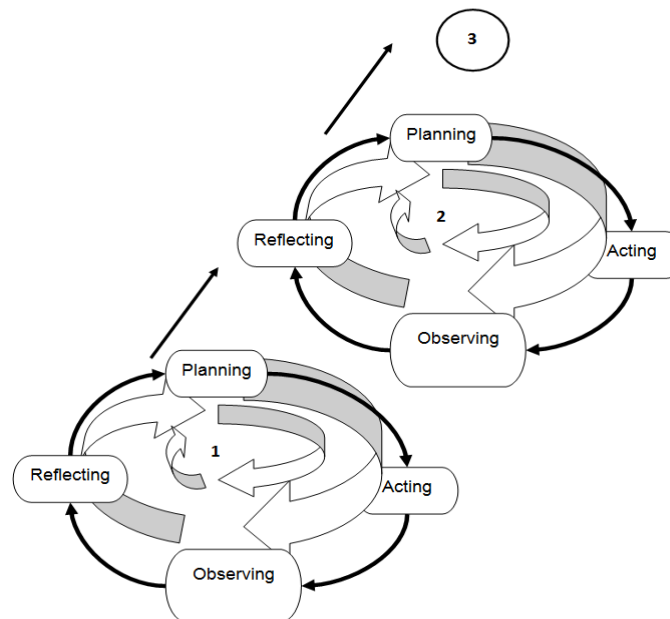


Fig. 6. Cyclical model of AR

3.2. Why Action Research?

Action Research is a valuable approach to explore how to change a situation or practice. So if research questions begin with 'How to' then Action Research may be the proper approach. It is also a valuable approach to build collaboration among various groups of practitioners, policy-makers, experts or community members. Some other major advantages of AR are presented below:

- Individuals become more effective when encouraged to examine and assess their own work and then consider ways to work differently.

- Individuals work best on problems they have identified for themselves.
- Individuals need time and space away from their daily routine to think deeply about their work.
- Mentors and mentees can provide help, support, and encouragement by working collaboratively.
- It can teach experts to identify problems systematically.
- It can build up a small community of research-oriented individuals at local level.

Gummeson (2000) presents some major characteristics of AR:

- 1) Action Researchers take action. They are not merely observing what happened; they are making it happen.
- 2) AR always aims at satisfying two goals: Solve a problem and contribute to science.
- 3) AR is interactive so it requires researcher and the client personnel co-operation.
- 4) AR tries to find complexity and develop holistic understanding during a project.
- 5) AR helps to explore how to change a situation or practice and it is fundamentally about change.
- 6) AR can include all types of data gathering methods such as quantitative and qualitative.

4. Suggested model for company processes

After studying the company and conducting interviews with its business experts, the treasury, training, distribution and sale, market research, inventory planning and control, warehousing, maintenance and purchasing processes have been identified. With classification of company processes into main processes and support processes, and selection of distribution and sales as the main process, inputs, outputs and key performance indicators (KPIs) of this process have been identified. This process should identify new customers, get orders from customers (sales representatives), distribute products based on customers' orders, get sales amounts from customers, and receive returned products and send them back to suppliers. Identified key performance indicators include conformity of real sales amount and sales quota, comparison between the amount of sales increase with the same amount in the last year, the time between receiving order sheets and product delivery to the customer, the product availability percentage with sales representative, representative coverage, accounts receivable period, and market share.

Fig. 7 shows the execution phases of this research. This is a dynamic model because it is a repetitive process that ends when the problem is solved. The model is similar to learning models used in Action Research projects, specially the learning cycle of Soft Systems Methodology.

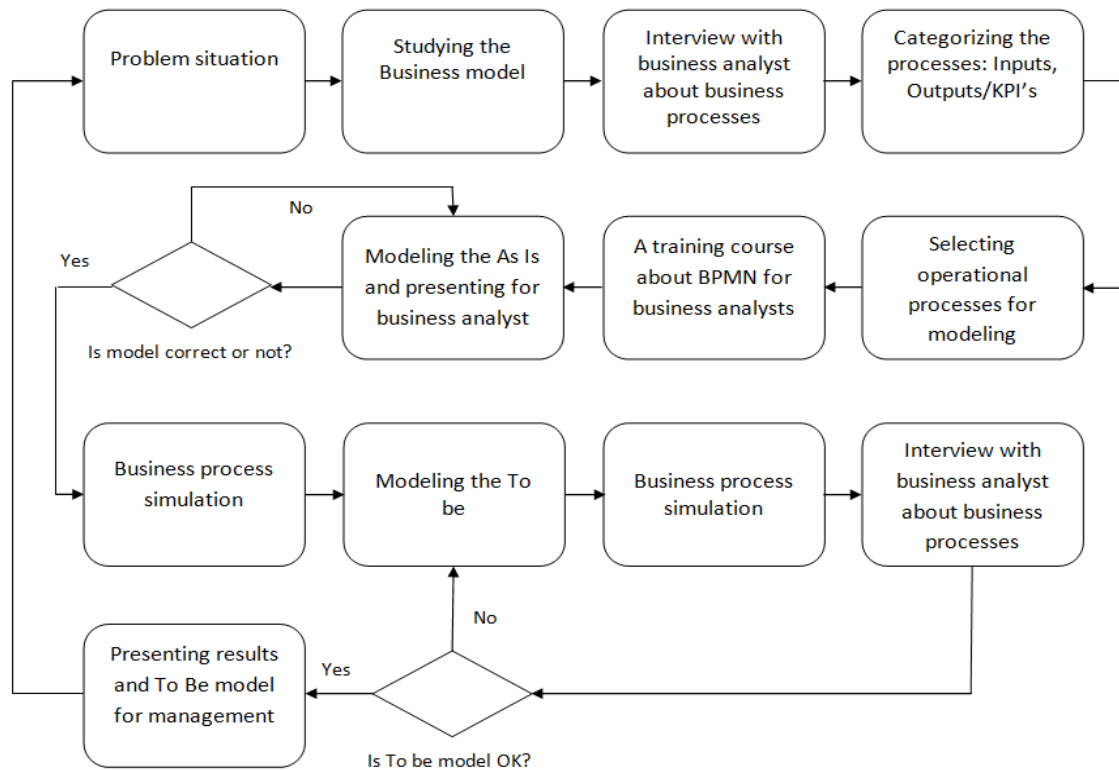


Fig. 7. Execution phases of Action Research

As shown in Fig 7, to bring business analysts and IT professionals together, a training course about BPMN was held for business analysts. This course helps business analysts to gain a better perception of IT professionals' duties and helps IT professionals to communicate more effectively with business analysts. In other words, In order to make the model operational in the next action, it was necessary to establish a relationship between the business analyst and the technology expert for implementing software solutions of the model. In this phase, activities were modeled with more details. This model helps a software expert a lot in its implementation (Kim, 2008). In this phase the conversion and mapping of BPMN graphical model to BPEL execution model is done using business process management software. The important point in this phase is the sufficient understanding of these two standards (OMG, 2008) and using structured modelling methods so that this mapping can be done correctly. After this mapping, the technology expert can complete implementing the model and making it executable using software tools and web services.

Once the sales and distribution process, and the sub-processes of getting orders, getting distribution agents, delivery, distribution, and their activities and tasks were identified, the current situation was modeled using BPMN standards.

Fig. 8 demonstrates a section of the current situation: the sub-process of getting orders from customers.

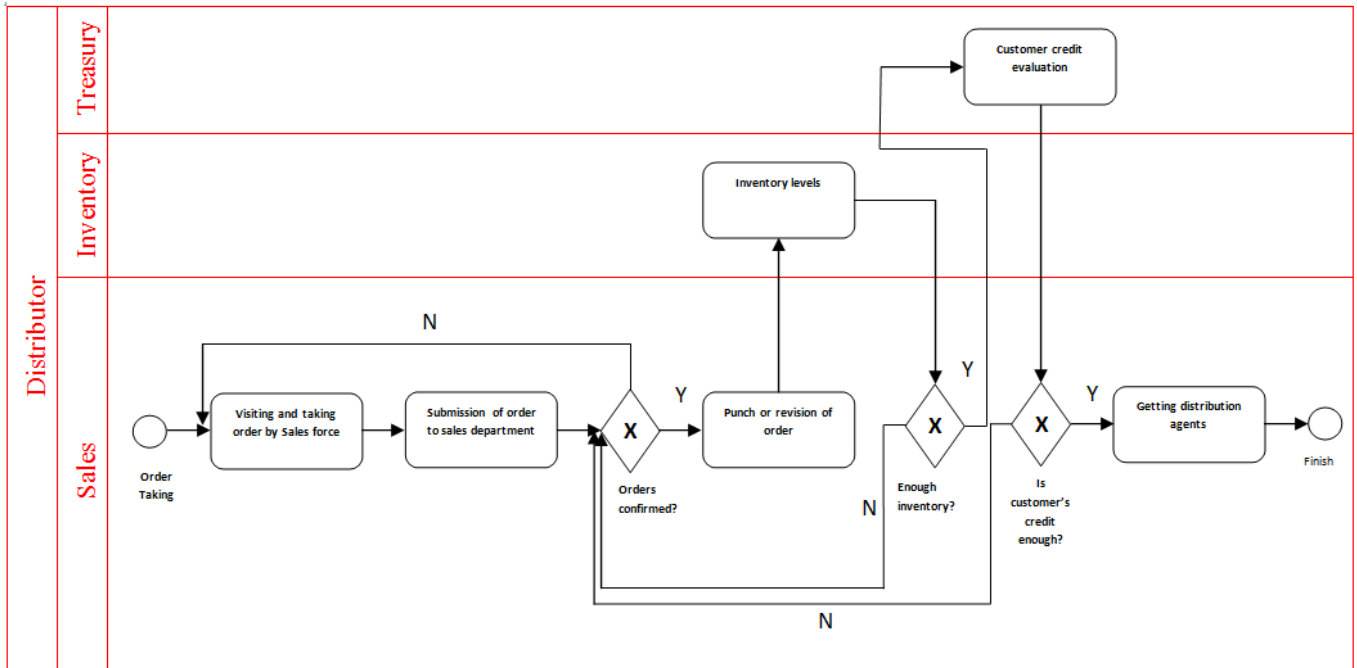


Fig. 8. The As Is model in sale order process

The standard uses BPMN to illustrate the current situation, since it is consistent with business rules and it's easy to use. Besides, this standard is tightly connected to BPEL as the model execution language. In this modelling method, the relation between business experts and technology experts will be established more easily. One of the modelling tools is UML which is mostly used for implementing software by technology experts and is not suitable for business experts to use by.

Once the model was presented to business experts and corrected, the proposed model for the future situation was created. In this new model, some changes in the current situation were proposed and done, enabling technology in order to improve the process. This is the second loop of the AR spiral. Sellers and distributors as the receivers of orders and settlement amount, made some changes in the manual operations of the sales and distribution process, from the starting point (getting orders) to the ending point (delivery of product and receipt of the settlement amount). Fig. 9 shows a section of this model in the future situation.

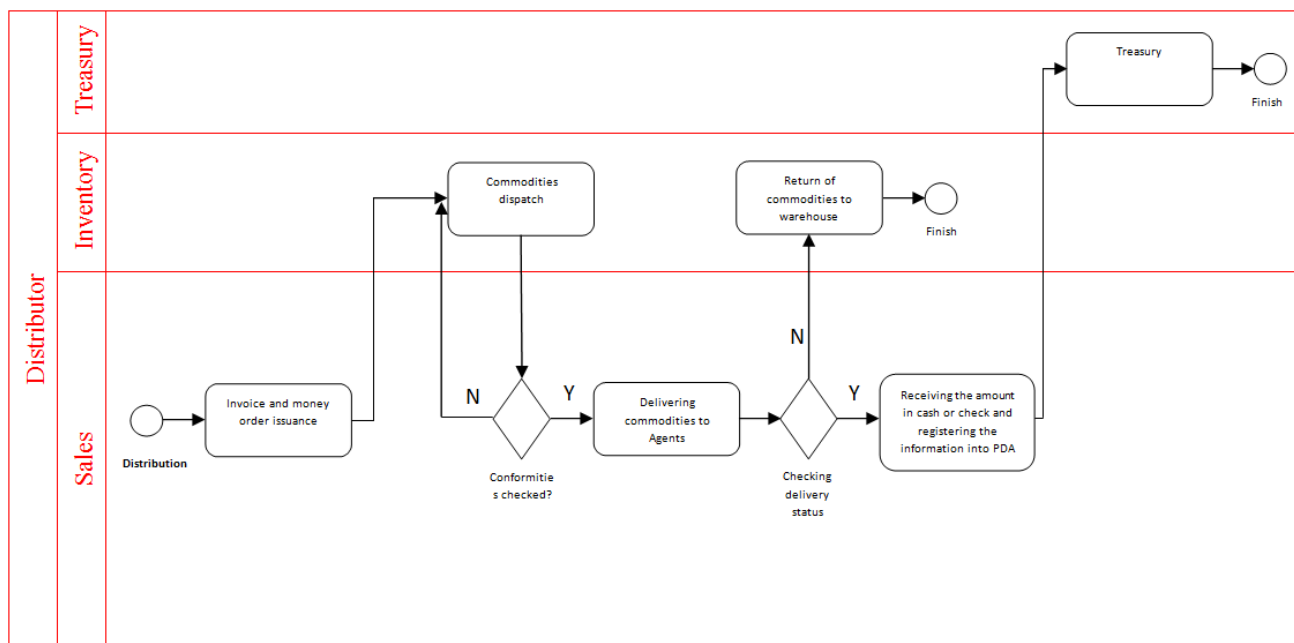


Fig. 9. The To-Be model in sale delivery process

Once the model was demonstrated in the As-Is and To-Be situations, the simulation of these situations and studying the results began. In order to simulate, first the resources used in this process were identified. The main indexes for model assessment were extracted. Two main indexes of cost and time were determined. Also human and hardware resources in the model were fixed. Then the indexes were quantified and assigned to resources in simulator software. Fig. 10 illustrates setting values to indexes in the model.

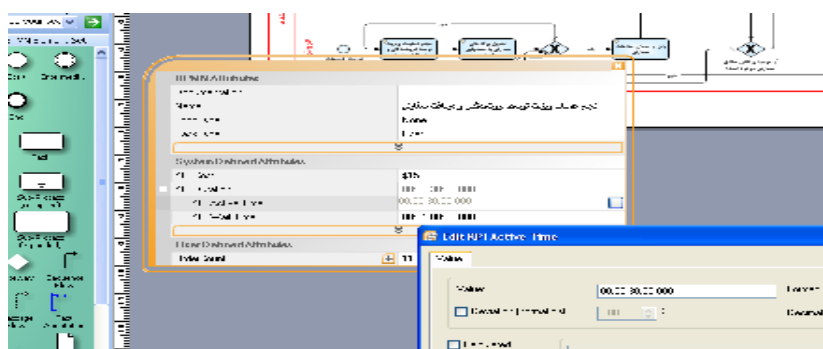


Fig. 10. Initiating the KPIs and resources value

Table 2 lists the used resources and their number in the simulation model.

Table 2 Resources and number of use in current and future state

Resource Type	Resource	No of use in current state	No of use in future state

Human resource	Sales man	12	12
Human resource	Sales operator	3	1
Human resource	Sales coordinator	2	1
Hardware	PDA devices	0	12
Hardware	Server for sales operation	0	1
Human resource	Financial operator	2	1
Hardware	PDA devices	0	3
Hardware	Server for financial operation	0	1

Once the simulation and required calculations were done and outputs were received in As-Is and To-Be situations, results were compared. As mentioned before, one of the main indexes considered was time. Active time and waiting time of each task in the model was determined and after the simulation, the time of human resources working was acquired (table 3).

Table 3 HR resource, active and wait time (Hour per week)

Resource	Active time in current state	Wait time in current state	Active time in future state	Wait time in future state
Sales man	282	240	432	90
Financial & Sales operator	150	25	60	10
Sales coordinator	68	12	34	6

Diagram 1 shows time and human resources cost used in As-Is and To-Be situations of sales and distribution processes. This diagram shows the percentage of active working time of human resources involved in this process and the cost of these resources for the company. The result of this simulation was improvement in human resources active time by 20 percent and decreasing in the cost of the human resource by 35 percent.

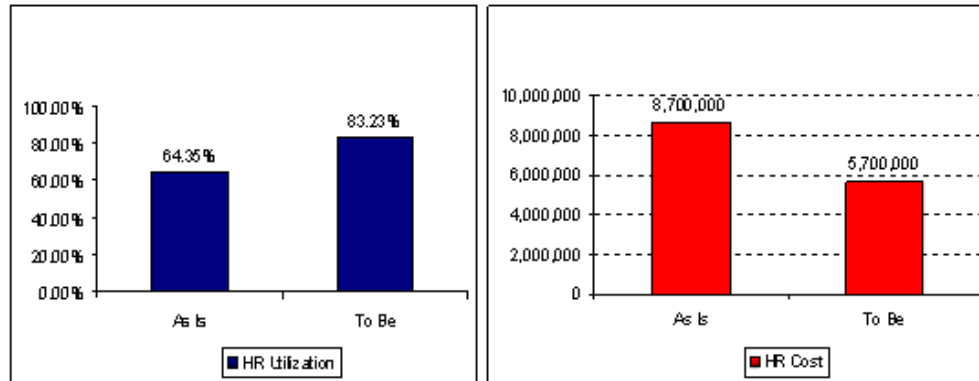


Diagram 1 Time usage and the human resource cost in the As-Is and To-Be situations

Presenting the changes modelling and the quantitative results to business experts and company managers, can stimulate the required motivation for correcting the process using the tools related to the technology and accepting its costs. The actions done so far were suitable for analysts and business experts. Comparing changes done in the process and their results and table 1 (which shows the rules presented in the business process reengineering), indicates that, the rules of deleting a task, combining tasks, suitable technology, parallelizing and decreasing the number of people involved in the process have been used directly in correcting sales and distribution process by using technology.

5. Conclusion

In order to gain business efficiency, competitive advantages and preparation for facing rapid market changes and competition with the rivals, organizations tend to use new enterprise information systems (Chiu *et al.*, 2006). The importance of correct identification of business processes, defining and correct use of them as well as improving them is among necessary matters in business.

It's clear that ever-increasing number of technologies and innovations resulted from them, can serve organizations, so that business processes of organizations will be improved more rapidly. Industries usually doubt in using these technologies and managers tend to use traditional and repetitive methods in organization's everyday activities. So, a gap is created between industries and technology. It's essential to provide them with understandable solutions and to make the results of using technology tangible for making agility in the organization. This article tried to show how to improve the business processes of the organization using two new modelling methods, namely BPMN and BPEL with an Action Research in a distribution company. Also, phases and steps of doing this research were presented. Besides, it showed how to quantify changes and to establish more relationships between the business analyst and the technology expert using business process modelling.

In this paper we have outlined our Action Research experiences in using BPMN during a process analysis and redesign project in a sales and distribution company. We found during our research that combination of BPMN and BPEL together enabled process participants to communicate easily with technical people. It is consistent with Gao's research (Gao, 2006) which asserts that business analysts and IT architects can have a better relationship by using BPMN and BPEL consequently.

We found BPMN models helped employees envision possible impacts of process changes on their daily work and also BPMN made it easy to communicate process redesign methods to senior management. In addition presenting simulation tools helped the business analyst to model and compare the results of changes to get acceptance of CEO for execute changes and making improvements in operational business processes. Usually in modelling approaches, the main objective of the project is documentation of the As-Is or To-Be situations but in this research, using BPMN and BPEL for automating models helped technical employees in IT department to execute some part of models in practice. This research shows that using graphical modelling methods with business process languages helps the business to achieve more flexibility in BPR. The result of this study was improvement in the human resources active time by 20 percent and decreasing in the cost of the human resource by 35 percent.

5.1. Limitations and future research directions

This study has some limitations that should be taken into consideration, especially due to the fact that this study is based on an Iranian distribution company. The suggested model might show different results in other companies or other countries. Subsequent research can explore these issues using a broader research sample.

Another important limitation is the fact that To-Be model is based on BPR in company's processes. BPR is a critical issue and previous researches have shown that as many as 60-80 percent of BPR projects have been unsuccessful (Dennis *et al.*, 2003). Therefore it requires a separate research.

Also human and hardware resources in the model were fixed. Therefore, any change in each of these resources may change research's outcomes.

Finally some researchers believe that there is a conceptual mismatch between BPMN and BPEL (Recker and Mendling, 2005) and this issue can affect some results of the study.

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